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An inverse problem for quantum systems in magnetic fields

A. Laestadius
(partially joint work with M. A. Csirik)

Department of Mathematical Sciences
Chalmers University of Technology / University of Gothenburg





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Density-functional theory (DFT)

For molecule-like systems, fix the pair interaction W and coupling $\kappa \geq 0$.

$$H_\kappa^v = -\sum_{i=1}^{N} \Delta_i + \sum_{i=1}^{N} v(\mathbf{x}_i) + \kappa W,$$

$$H_\kappa^v \Psi = E\Psi, \qquad E = \inf_{\|\psi\|_2=1} \langle \psi, H_\kappa^v \psi \rangle.$$

Full ground state

$$\Psi(\mathbf{x}_1, \ldots, \mathbf{x}_N)$$

DFT →

Particle density

$$\rho_\Psi(\mathbf{x}) = N \int_{\mathbb{R}^{3(N-1)}} |\Psi(\mathbf{x}, \mathbf{X})|^2 \, d\mathbf{X}$$

Spin omitted. $\mathbf{X} = (\mathbf{x}_2, \ldots, \mathbf{x}_N)$. $\int \rho_\Psi = N$.

Hohenberg–Kohn (HK): Ground-state density determines v up to a constant (under technical assumptions). This is the inverse-uniqueness foundation of DFT.

P. Hohenberg and W. Kohn, *Phys. Rev.* **136**, B864 (1964).

The scalar HK argument

Ground states Ψ_i of $H_i = H_\kappa^{v_i}$ have the same ρ.

Variational principle

$$E_1 \leq E_2 + \int (v_1 - v_2)\rho,$$

$$E_2 \leq E_1 - \int (v_1 - v_2)\rho.$$

Both are equalities. Thus Ψ_2 is a common ground state.
Subtracting its eigenvalue equations gives

$$\Big[\sum_{j=1}^{N} (v_1 - v_2)(\mathbf{x}_j) - (E_1 - E_2)\Big]\Psi_2 = 0.$$

Garrigue's strong UCP[1]

In $d = 3$, take $v_i, w \in L^p_{\mathrm{loc}}$, $p > 2$.

Every nonzero H^2_{loc} eigenfunction is nonzero a.e., independently of N (via a Carleman estimate).

Coulomb potentials are included.

$v_1 - v_2 = (E_1 - E_2)/N$ a.e. Ground-state degeneracy is allowed.

T. C. Corso & **A. L.**, Quantitative HK and stable potential recovery in 1D (accepted in *J. Spectr. Theory*, Oct. 2026).

[1] L. Garrigue, *Math. Phys. Anal. Geom.* **21**, 27 (2018).

Current-density-functional theory (CDFT)

N spinless fermions, with fixed interaction W and coupling $\kappa \geq 0$.

$$H_\kappa^{v,\mathbf{A}} = \sum_{i=1}^{N} \left[(-i\nabla_i + \mathbf{A}(\mathbf{x}_i))^2 + v(\mathbf{x}_i) \right] + \kappa W.$$

Paramagnetic current

$$\mathbf{j}_\Psi^{\mathrm{p}} = N \operatorname{Im} \int_{\mathbb{R}^{3(N-1)}} \overline{\Psi} \nabla_{\mathbf{x}} \Psi \, d\mathbf{X}$$

Determined by the state. Gauge dependent.

Total or physical current

$$\mathbf{j}_\Psi^{\mathbf{A}} = \mathbf{j}_\Psi^{\mathrm{p}} + \rho_\Psi \mathbf{A}$$

Depends on the state and $\mathbf{A}$. Gauge invariant.

The total-current HK conjecture[1] $\quad (\rho_1, \mathbf{j}_1) = (\rho_2, \mathbf{j}_2) \implies v_1 - v_2 = \text{constant}, \ \mathbf{A}_1 - \mathbf{A}_2 = \nabla\eta.$

[1] G. Diener, *J. Phys.: Condens. Matter* **3**, 9417 (1991).

Starting point with M. Benedicks

2014 Which information determines what?

For a nondegenerate ground state in a fixed gauge,

$$(\rho, \mathbf{j}^{\mathrm{p}}) \text{ determines } \Psi \text{ up to phase,} \qquad \Psi \text{ need not determine } (v, \mathbf{A}).$$

Infinitely many potentials can share a ground state.
For **one particle**, $(\rho, \mathbf{j})$ determines the potentials up to the usual freedoms, under suitable regularity assumptions.

2015 Uniqueness and minimization are different questions

An HK expectation-value functional of $(\rho, \mathbf{j})$ can attain the ground-state energy at the **wrong density–current pair**, even for one particle.

A. L. and M. Benedicks, *Int. J. Quantum Chem.* **114**, 782 (2014) and *Phys. Rev. A* **91**, 032508 (2015).

Continuing with Oslo colleagues and Markus Penz

2018 With Erik Tellgren

Equal $(\rho, \mathbf{j}^{\mathrm{P}})$ implies a common ground state, even with degeneracy.
A. L. and E. I. Tellgren, *Phys. Rev. A* **97**, 022514 (2018).

2019 With Tellgren, Penz, Ruggenthaler, Kvaal and Helgaker

Compatible function spaces and differentiable, regularized functionals for Kohn–Sham theory with paramagnetic currents.
A. L., E. I. Tellgren, M. Penz, M. Ruggenthaler, S. Kvaal, and T. Helgaker, *J. Chem. Theory Comput.* **15**, 4003 (2019).

2021 With Markus Penz and Erik Tellgren

Diener's total-current variational construction fails in general.
A. L., M. Penz, and E. I. Tellgren, *J. Phys.: Condens. Matter* **33**, 295504 (2021).

2023 With Penz, Tellgren, Csirik and Ruggenthaler

A synthesis of magnetic density–potential mappings and the open questions.
M. Penz, E. I. Tellgren, M. A. Csirik, M. Ruggenthaler, and **A. L.**, *ACS Phys. Chem. Au* **3**, 492 (2023).

Counterexample for HK with total current

Idea: Take the complex conjugate of the eigenvalue equation.

$$H^{v,\mathbf{A}}\Psi = E\Psi \quad \overset{\text{c.c.}}{\Longleftrightarrow} \quad H^{v,-\mathbf{A}}\overline{\Psi} = E\overline{\Psi}.$$

Original system

Ψ ground state of $H^{v,\mathbf{A}}$

$(\rho, \mathbf{j}, \mathbf{B})$

$\overset{\text{c.c.}}{\Longleftrightarrow}$

Reversed system

$\overline{\Psi}$ ground state of $H^{v,-\mathbf{A}}$

$(\rho, -\mathbf{j}, -\mathbf{B})$

It suffices to find a nondegenerate ground state with

$$\mathbf{j} = 0 \qquad \text{and} \qquad \mathbf{B} \neq 0.$$

The scalar potential, interaction, and spectrum are unchanged.

Two orbitals can cancel their currents pointwise

1. Choose two orthonormal orbitals $\kappa = 0$, cylindrical (r, θ, z)

$$\phi_0(\mathbf{x}) = Cf(r)\chi(z) \quad (m = 0), \qquad \phi_-(\mathbf{x}) = q(r)e^{-i\theta}\phi_0(\mathbf{x}) \quad (m = -1).$$

Choose f, q, χ, C so both orbitals have norm one, with $f, q, \chi > 0$ for $r > 0$. The angular phases give orthogonality.

2. Build the two-fermion Slater determinant

$$\Psi(\mathbf{x}_1, \mathbf{x}_2) = \frac{\phi_0(\mathbf{x}_1)\phi_-(\mathbf{x}_2) - \phi_-(\mathbf{x}_1)\phi_0(\mathbf{x}_2)}{\sqrt{2}}.$$

Exchanging the particles changes the sign. The factor $1/\sqrt{2}$ normalizes the state.

3. Choose $\mathbf{A}$ to cancel the total current

$$\rho = |\phi_0|^2 + |\phi_-|^2 = (1 + q^2)|\phi_0|^2,$$

$$\mathbf{j}^{\mathrm{p}} = -\frac{|\phi_-|^2}{r}\mathbf{e}_\theta.$$

Only ϕ_- contributes to $\mathbf{j}^{\mathrm{p}}$, through its angular phase.

$$\mathbf{A} = -\frac{\mathbf{j}^{\mathrm{p}}}{\rho} = \frac{q^2}{r(1 + q^2)}\mathbf{e}_\theta$$

$$\mathbf{j} = \mathbf{j}^{\mathrm{p}} + \rho\mathbf{A} = 0 \qquad (r > 0).$$

Next, choose the profiles and scalar potential to obtain a smooth nonzero field and a nondegenerate ground state.

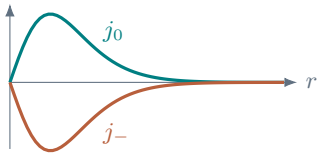
An explicit example with a Gaussian density

$$\phi_0 = \sqrt{\frac{2}{\pi}} \left(\frac{c}{\pi}\right)^{1/4} \frac{e^{-(r^2+cz^2)/2}}{\sqrt{1+tr^2}}, \qquad \phi_- = \sqrt{t}\,(x-iy)\phi_0.$$

$$f(r) = \frac{e^{-r^2/2}}{\sqrt{1+tr^2}}, \qquad q(r) = \sqrt{t}\,r, \qquad \chi(z) = \left(\frac{c}{\pi}\right)^{1/4} e^{-cz^2/2}.$$

Choose $c > 1$. Normalization fixes $t \approx 1.63919$ through $\displaystyle\int_0^\infty \frac{e^{-s}}{1+ts}\,ds = \frac{1}{2}$.

$$\mathbf{A} = \frac{t(-y, x, 0)}{1+tr^2}, \qquad \mathbf{B} = \frac{2t}{(1+tr^2)^2}\mathbf{e}_z.$$



Azimuthal orbital currents at fixed z, up to a common positive factor

$$j_0 = -j_- = \rho\, tr/(1+tr^2)^2.$$

Identical densities for $+\mathbf{A}$ and $-\mathbf{A}$

$$\rho = \frac{2}{\pi}\sqrt{\frac{c}{\pi}}\, e^{-r^2-cz^2}, \qquad \mathbf{j} = 0.$$

$\mathbf{A}$ is smooth and bounded. The scalar potential is a confining oscillator plus a bounded correction.

The essential step is proving ground-state ordering

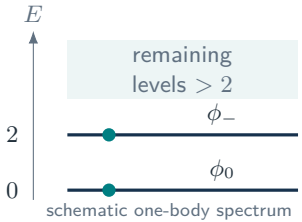
$$H^{v,\mathbf{A}}_{\kappa=0} = h_1^{v,\mathbf{A}} + h_2^{v,\mathbf{A}}, \qquad h^{v,\mathbf{A}} = (-i\nabla + \mathbf{A})^2 + v.$$

In the angular sector $e^{im\theta}$, $\quad h^{v,\mathbf{A}}\big|_m = \mathcal{L}_m - \partial_z^2 + c^2 z^2 - c.$

$$\mathcal{L}_m = -\partial_r^2 - \frac{1}{r}\partial_r + \left(\frac{m}{r} + \frac{tr}{1+tr^2}\right)^2 + r^2 - \frac{2}{1+tr^2} - \frac{2t}{(1+tr^2)^2}.$$

$\mathcal{L}_m$ acts on $L^2((0,\infty), r\,dr)$. The radial and axial energies add.

- $\mathcal{L}_0 f = 0$, $\mathcal{L}_{-1}(rf) = 2rf$ are simple minima.
- Excited $m = 0$ levels have $E \geq 4$.
- Other angular sectors have $E > 2$.
- Axial excitations have $E \geq 2c > 2$.



schematic one-body spectrum

Filling the two lowest orbitals gives a unique two-fermion ground state.

Nonuniqueness persists for smooth bounded potentials

Theorem: the bounded construction (A. L. & M. A. Csirik)[1]

There exist real $v \in C^\infty \cap L^\infty$ and $\mathbf{A} \in (C^\infty \cap L^\infty)^3$ on $\mathbb{R}^3$ with $\nabla \times \mathbf{A} \neq 0$ such that $H^{v,\pm\mathbf{A}}$ have isolated, nondegenerate ground states and

$$\rho_+ = \rho_- > 0, \qquad \mathbf{j}_+ = \mathbf{j}_- = 0 \qquad (\kappa = 0).$$

Both potentials belong to the standard spaces

$$\mathcal{V} = L^{3/2}(\mathbb{R}^3) + L^\infty(\mathbb{R}^3), \qquad \mathcal{A} = \big(L^3(\mathbb{R}^3) + L^\infty(\mathbb{R}^3)\big)^3.$$

Exponential radial decay, a bounded axial well, and the same factorization strategy.

Explicit formulas depend on a normalization parameter whose existence follows from the intermediate value theorem.

[1] **A. L.** & M. A. Csirik, *Counterexamples to the total-current Hohenberg–Kohn conjecture*. arXiv:2610.27294 (2026), Theorem 3.1(ii).

The same density and total current can hide opposite fields

Two nondegenerate ground states can have

$$\rho_+ = \rho_-, \qquad \mathbf{j}_+ = \mathbf{j}_- = 0, \qquad \mathbf{B}_+ = -\mathbf{B}_- \neq 0.$$

The scalar potential is the same in both systems.

Noninteracting systems An explicit Gaussian example, and a construction with smooth bounded $(v, \mathbf{A}) \in \mathcal{V} \times \mathcal{A}$.

A brief remark on interaction

A counterexample with Coulomb repulsion can also be constructed using monopole connections **outside the standard potential spaces**.

These constructions do not settle the interacting problem within those spaces.

Thank you

New results with Mihály A. Csirik

A. L. and M. A. Csirik
Counterexamples to the total-current Hohenberg–Kohn conjecture
arXiv:2610.27294 (2026).

Acknowledgements

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Penz, Tellgren, Csirik, Ruggenthaler, and **A. L.** (2023)
The structure of the density-potential mapping. Part II. Including magnetic fields

Further references are collected in the backup slides.

REGAL repository